

SDOF BEAM SOLVER USING FUNCTIONS

1. Code

```
import numpy as np
import matplotlib.pyplot as plt
def sdof_response(m, k, c, x0, v0, t):

    omega_n = np.sqrt(k / m)

    zeta = c / (2 * np.sqrt(k * m))

    if zeta < 1:

        omega_d = omega_n * np.sqrt(1 - zeta**2)
        A = x0
        B = (v0 + zeta * omega_n * x0) / omega_d

        x = np.exp(-zeta * omega_n * t) * (
            A * np.cos(omega_d * t)
            + B * np.sin(omega_d * t)
        )

    else:

        x = np.zeros_like(t)

    return x

# INPUT PARAMETERS
m = 1
k = 100
x0 = 0.1
v0 = 0

t = np.linspace(0, 5, 1000)

# Three damping cases
case1 = sdof_response(m, k, 0, x0, v0, t)
case2 = sdof_response(m, k, 1, x0, v0, t)
case3 = sdof_response(m, k, 10, x0, v0, t)

# Plot
plt.figure(figsize=(8,5))
plt.plot(t, case1, label="c = 0")
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plt.plot(t, case2, label="c = 1")
plt.plot(t, case3, label="c = 10")
plt.xlabel("Time (s)")
plt.ylabel("Displacement (m)")
plt.title("Effect of Damping on Free Vibration")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

print("\nEffect of Stiffness")
k_values = [50, 100, 200, 500]
plt.figure(figsize=(8,5))

for k in k_values:

    response = sdof_response(m, k, 1, x0, v0, t)
    omega_n = np.sqrt(k/m)
    print(f"k = {k:3d} N/m    "
          f"Natural Frequency = {omega_n:.3f} rad/s")

    plt.plot(t, response, label=f"k = {k}")

plt.xlabel("Time (s)")
plt.ylabel("Displacement (m)")
plt.title("Effect of Stiffness")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

print("\nEffect of Mass")
m_values = [0.5, 1, 2, 5]
plt.figure(figsize=(8,5))

k = 100

for m in m_values:

    response = sdof_response(m, k, 1, x0, v0, t)

    omega_n = np.sqrt(k/m)

    print(f"m = {m:3.1f} kg    "
          f"Natural Frequency = {omega_n:.3f} rad/s")

```

```

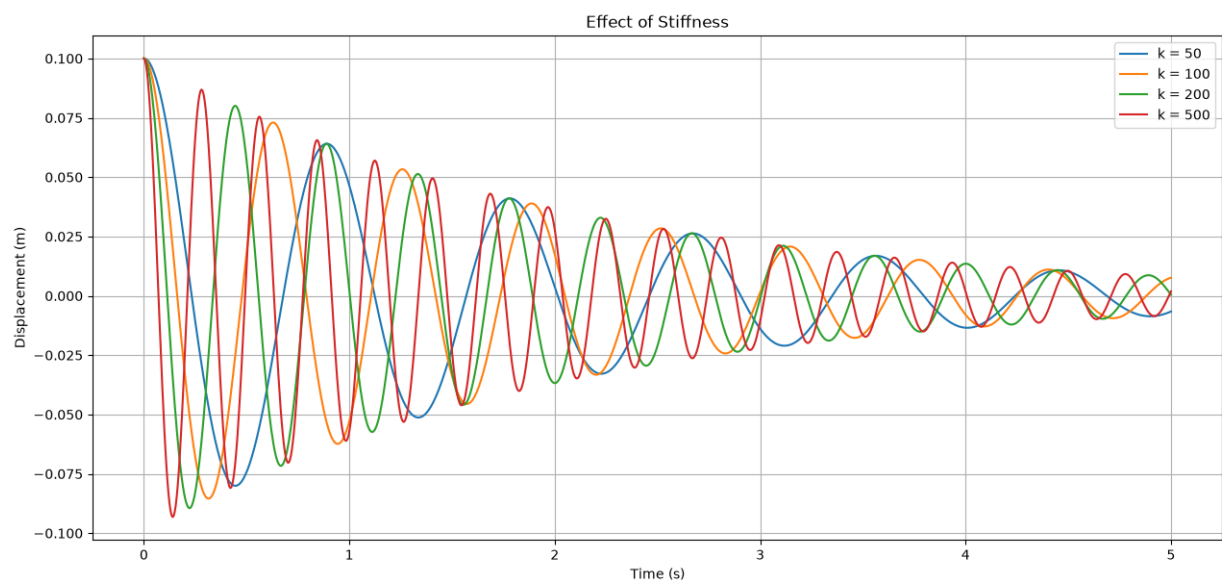
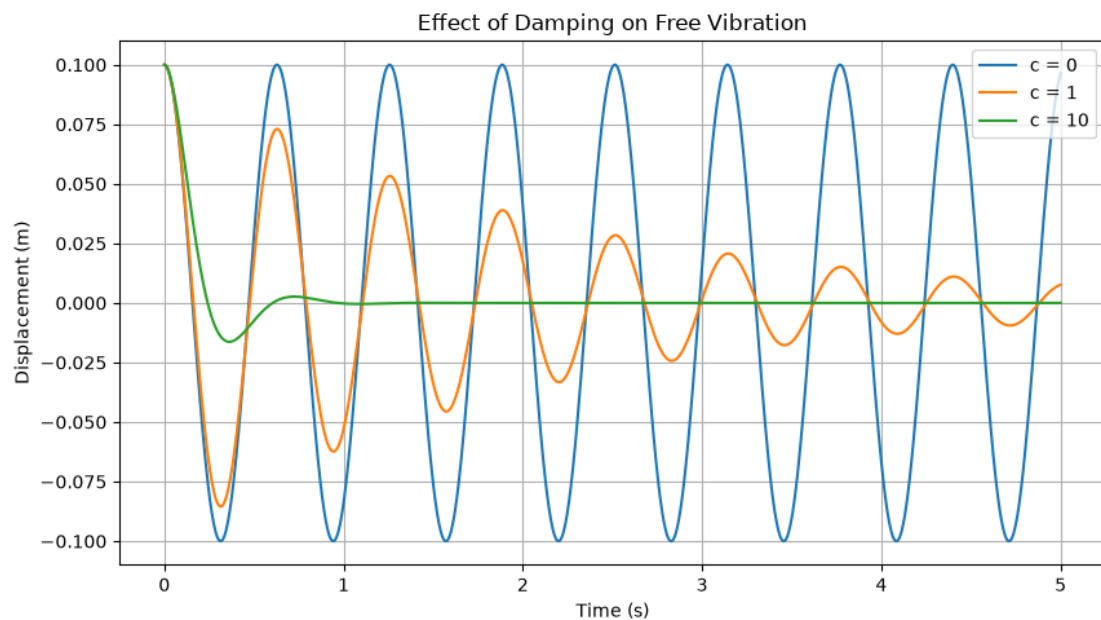
plt.plot(t, response, label=f"m = {m}")

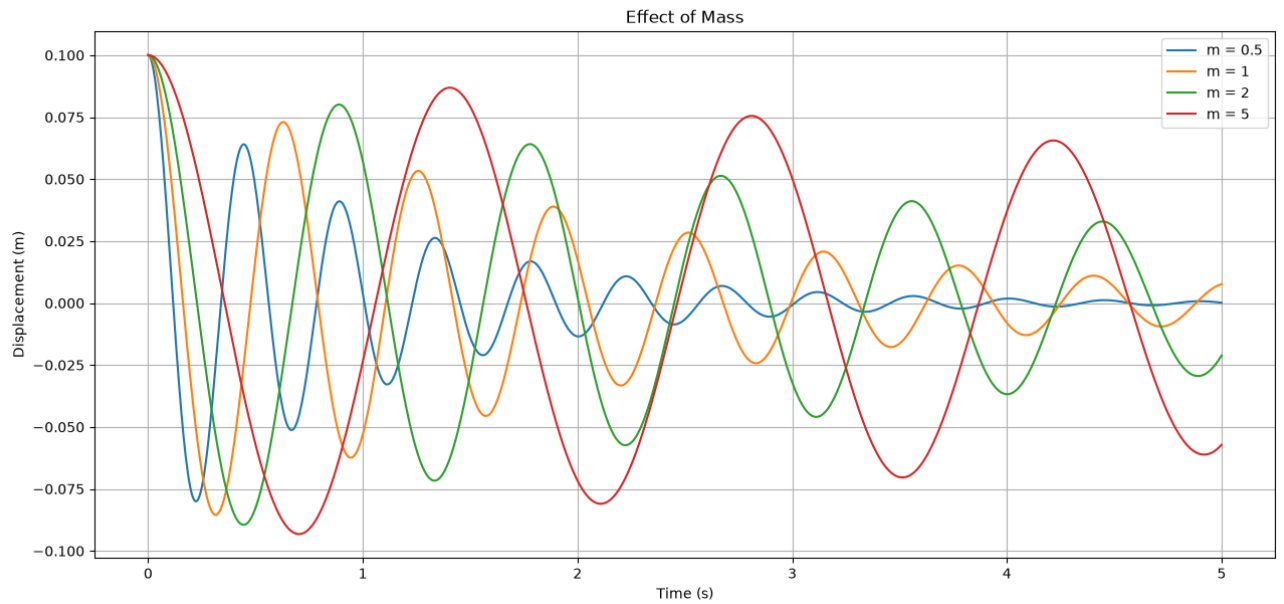
plt.xlabel("Time (s)")
plt.ylabel("Displacement (m)")
plt.title("Effect of Mass")

plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```

2. Plots





Effect of Stiffness

k = 50 N/m	Natural Frequency = 7.071 rad/s
k = 100 N/m	Natural Frequency = 10.000 rad/s
k = 200 N/m	Natural Frequency = 14.142 rad/s
k = 500 N/m	Natural Frequency = 22.361 rad/s

Effect of Mass

m = 0.5 kg	Natural Frequency = 14.142 rad/s
m = 1.0 kg	Natural Frequency = 10.000 rad/s
m = 2.0 kg	Natural Frequency = 7.071 rad/s
m = 5.0 kg	Natural Frequency = 4.472 rad/s

3. Why does increasing mass reduce natural frequency?

The natural frequency is inversely proportional to the square root of the mass,

$$\omega_n = \sqrt{\frac{k}{m}}$$

An increase in mass increases the inertia of the system, making it more resistant to acceleration. As a result, the oscillation period increases, and the natural frequency decreases.